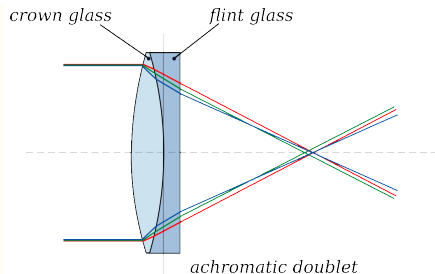
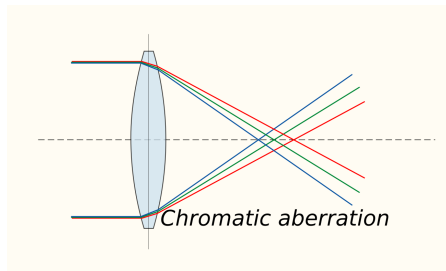


Chromaticity

Chromatic effects



- ▶ Chroma = color
- ▶ In our case, “color” means “energy”
- ▶ Since we have dispersion (generated by dipoles), particles with different momenta have different trajectories
- ▶ Focused differently by quads
- ▶ Tune shifts for particle with nonzero $\Delta p/p$

Chromaticity and tune shift

$$K = \frac{eg}{p}, \Delta K = \frac{dK}{dp} \Delta p = -\frac{eg}{p_0} \frac{\Delta p}{p_0} = -K_0 \frac{\Delta p}{p_0},$$

where g is the quadrupole gradient, p_0 is the reference momentum, K_0 is the quadrupole strength for on-momentum particles

- ▶ It can be shown (see CERN'94 paper for details, pp. 75–76) that $\Delta Q = \frac{1}{4\pi} \oint \beta(s) \Delta K(s) ds$, where ΔQ is the tune shift corresponding to the quadrupole error ΔK
- ▶ Substitute our expression for ΔK into this formula,

$$\Delta Q = -\frac{1}{4\pi} \oint \beta(s) K(s) ds \frac{\Delta p}{p_0} = \xi \frac{\Delta p}{p_0}, \quad \xi = -\frac{1}{4\pi} \oint \beta(s) K(s) ds$$

- ▶ Here ξ – chromaticity – is the coefficient between ΔQ (tune shift) and $\Delta p/p_0$

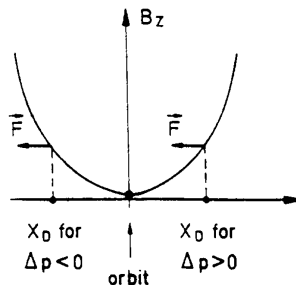
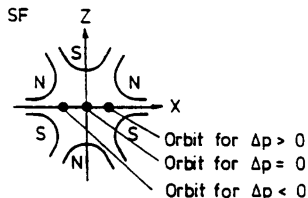
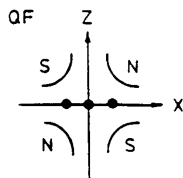
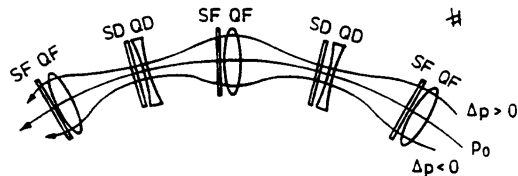
Natural chromaticity, chromaticity correction

$$\xi = -\frac{1}{4\pi} \oint \beta(s) K(s) ds$$

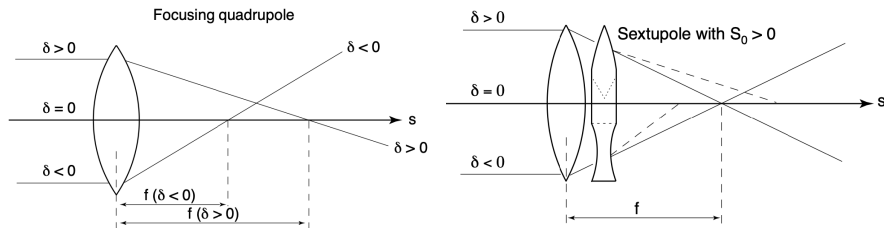
- ▶ For a linear lattice $\xi < 0$
- ▶ Chromaticity arising from the linear lattice is called “natural” or “linear”
- ▶ Could be large, especially where β is large (e.g., final focus)
- ▶ Leads to large tune shift $\Rightarrow$ unwanted resonance crossings (e.g., half- or third-integer ones)
- ▶ Chromaticity correction is in order

Sextupoles

- ▶ Sextupole: the first nonlinear element we encounter in the transverse direction
- ▶ Placement: where $D(s) \neq 0$
- ▶ In FODO cells: SF by QF and SD by QD



Chromaticity correction



With sextupoles included, the lattice chromaticity can be found to be

$$\xi_x = -\frac{1}{4\pi} \oint \beta_x [K_x(s) - S(s)D(s)] ds,$$

$$\xi_y = -\frac{1}{4\pi} \oint \beta_y [K_y(s) + S(s)D(s)] ds$$

FODO cell

$$S_F = -\frac{B''(F)}{B\rho}L_{SF}, S_D = -\frac{B''(D)}{B\rho}L_{SD},$$

where $S_{F,D}$ are the integrated sextupole strengths at QF and QD, $L_{SF,SD}$ and $B''(F,D)$ are the length and sextupole field strength at QF and QD.

For zero chromaticity we need

$$S_F = \frac{1}{2f^2\theta} \frac{\sin(\frac{\mu}{2})}{1 + \frac{1}{2}\sin(\frac{\mu}{2})}$$

$$S_D = \frac{1}{2f^2\theta} \frac{\sin(\frac{\mu}{2})}{1 - \frac{1}{2}\sin(\frac{\mu}{2})}$$

Here f is the focal distance, θ is the bending angle per *half-cell*, μ is the phase advance per *cell*. Note the asymmetry in the formulas.

Derivation is left as an exercise

MAD-X could help us find the right values [\[demo\]](#)

FODO: effect on chromaticity

The chromaticity change due to the sextupoles is

$$\Delta\xi_x = \frac{N}{2\pi}(\beta_{\max}D_{\max}S_F + \beta_{\min}D_{\min}S_D),$$

$$\Delta\xi_y = -\frac{N}{2\pi}(\beta_{\min}D_{\max}S_F + \beta_{\max}D_{\min}S_D),$$

Note. In order to minimize sextupole strengths, the chromatic sextupoles should be located near quadrupoles, where $\beta_x D_x$ and $\beta_y D_x$ are at maximum.

Note. Unfortunately, the sextupoles inevitably introduce intrinsically nonlinear aberrations, and we have taken our first step away from a design based upon purely linear dynamics.

Note. For a complex design, two families of sextupoles may not be sufficient to correct the higher-order chromatic effects.